

## 8.6 The Cayley–Hamilton Theorem

An operator on an  $n$ -dimensional space now carries two monic polynomials. The characteristic polynomial  $p_T$  comes from determinants and has degree exactly  $n$ . The minimal polynomial  $m_T$  comes from the ideal of polynomials that kill  $T$ , and the previous section showed that it has the same roots as  $p_T$ . Chapter 6 stated, without proof, the fact that ties the two together: every square matrix satisfies its own characteristic polynomial. Chapter 6 also warned against a one-line fake proof of that statement. This section gives a real proof, built on the adjugate identity of Chapter 6 read over  $F[x]$ . It then draws the consequences:  $m_T$  divides  $p_T$ , inverses are polynomials, and high powers of a matrix reduce to low ones.

### 8.6.1 The statement, and a proof that is not one

Chapter 5 proved that every operator on an  $n$ -dimensional space is killed by some non-zero polynomial of degree at most  $n^2$  (Theorem 5.7.2). The bound came from counting:  $\mathcal{L}(V)$  has dimension  $n^2$ . Counting cannot do better, yet in every example so far a polynomial of degree at most  $n$  sufficed. The characteristic polynomial always has degree  $n$ , and it is the natural candidate.

#### Theorem 8.6.1. Cayley–Hamilton Theorem

Let  $n \geq 1$  and  $\mathbf{A} \in M_n(F)$ . Then

$$p_{\mathbf{A}}(\mathbf{A}) = 0.$$

Likewise, if  $V$  is a finite-dimensional vector space over  $F$  with  $\dim V \geq 1$  and  $T \in \mathcal{L}(V)$ , then  $p_T(T) = 0$ .

Here  $p_{\mathbf{A}}(\mathbf{A})$  means what it always means (Definition 0.9.14): if  $p_{\mathbf{A}} = x^n + c_{n-1}x^{n-1} + \cdots + c_1x + c_0$ , then

$$p_{\mathbf{A}}(\mathbf{A}) = \mathbf{A}^n + c_{n-1}\mathbf{A}^{n-1} + \cdots + c_1\mathbf{A} + c_0\mathbf{I}_n,$$

an  $n \times n$  matrix, with the constant term becoming  $c_0\mathbf{I}_n$ .

For  $n = 2$  the theorem can be checked by hand. With  $\mathbf{A} = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ ,  $p_{\mathbf{A}} = x^2 - (\operatorname{tr} \mathbf{A})x + \det \mathbf{A}$ , and Exercise (C2: Nilpotent  $2 \times 2$  matrices) (b) multiplied out

$$\mathbf{A}^2 - (a + d)\mathbf{A} + (ad - bc)\mathbf{I}_2 = 0.$$

For  $n = 3$ , the same brute force already involves cubic expressions in nine entries, and for general  $n$  it is hopeless. We need an argument. There is a tempting one, and we name it before it can do any harm.

**Warning. The fake proof of Cayley–Hamilton, again.** “Since  $p_{\mathbf{A}}(x) = \det(x\mathbf{I} - \mathbf{A})$ , substitute  $x = \mathbf{A}$  to get  $p_{\mathbf{A}}(\mathbf{A}) = \det(\mathbf{A}\mathbf{I} - \mathbf{A}) = \det 0 = 0$ .” Chapter 6 (the warning at the end of its §10) explained why this is not a proof. The left side  $p_{\mathbf{A}}(\mathbf{A})$  is a **matrix**, while  $\det(\cdots)$  is a **scalar**, so the equation compares objects of different kinds. And substituting  $\mathbf{A}$  for  $x$  **inside the entries** of  $x\mathbf{I} - \mathbf{A}$  is not what substituting into a polynomial means: the entry  $x - a_{11}$  would have to become the matrix  $\mathbf{A} - a_{11}\mathbf{I}$ , and the determinant of a matrix whose entries are matrices is undefined. The same “argument” applied to  $q(x) = \operatorname{tr}(x\mathbf{I} - \mathbf{A}) = nx - \operatorname{tr} \mathbf{A}$  would prove  $q(\mathbf{A}) = 0$ , which is false for  $\mathbf{A} = \operatorname{diag}(1, 0)$ : there

$q(\mathbf{A}) = 2\mathbf{A} - \mathbf{I} = \text{diag}(1, -1)$ . Any correct proof must keep  $x$  a polynomial variable until the very end, and only then put a matrix in its place.

## 8.6.2 A proof through the adjugate

The adjugate identity of Chapter 6 says  $\mathbf{M} \text{adj } \mathbf{M} = (\det \mathbf{M}) \mathbf{I}$  for every square matrix  $\mathbf{M}$ , and the remark after Theorem 6.6.2 notes that its proof works over any commutative ring. Over  $F[x]$ , with  $\mathbf{M} = x\mathbf{I} - \mathbf{A}$ , it reads

$$(x\mathbf{I} - \mathbf{A}) \text{adj}(x\mathbf{I} - \mathbf{A}) = p_{\mathbf{A}}(x) \mathbf{I}. \quad (1)$$

This is an identity between matrices of **polynomials**, and it contains  $p_{\mathbf{A}}$  on the right. The proof extracts  $p_{\mathbf{A}}(\mathbf{A})$  from it legally.

*Idea.* Try  $n = 2$  first. By Exercise (C3: An adjugate over  $F[x]$ ) (b) (done there over  $\mathbb{R}$ , but the computation works over any field),  $\text{adj}(x\mathbf{I} - \mathbf{A}) = x\mathbf{I} - \text{adj } \mathbf{A}$ , a polynomial in  $x$  with **matrix coefficients**  $\mathbf{B}_1 = \mathbf{I}$  and  $\mathbf{B}_0 = -\text{adj } \mathbf{A}$ . Expanding the left side of (1),

$$(x\mathbf{I} - \mathbf{A})(x\mathbf{I} - \text{adj } \mathbf{A}) = x^2 \mathbf{I} - x(\mathbf{A} + \text{adj } \mathbf{A}) + \mathbf{A} \text{adj } \mathbf{A}.$$

Comparing the coefficients of  $x^2$ ,  $x$  and 1 with those of  $(x^2 - (\text{tr } \mathbf{A})x + \det \mathbf{A})\mathbf{I}$  gives three matrix equations:

$$\mathbf{I} = \mathbf{I}, \quad -(\mathbf{A} + \text{adj } \mathbf{A}) = -(\text{tr } \mathbf{A}) \mathbf{I}, \quad \mathbf{A} \text{adj } \mathbf{A} = (\det \mathbf{A}) \mathbf{I}.$$

Now multiply the first by  $\mathbf{A}^2$ , the second by  $\mathbf{A}$ , the third by  $\mathbf{I}$ , and add. The right sides add up to  $\mathbf{A}^2 - (\text{tr } \mathbf{A})\mathbf{A} + (\det \mathbf{A})\mathbf{I} = p_{\mathbf{A}}(\mathbf{A})$ . The left sides add up to  $\mathbf{A}^2 - \mathbf{A}^2 - \mathbf{A} \text{adj } \mathbf{A} + \mathbf{A} \text{adj } \mathbf{A}$ , in which every term cancels against its neighbor. So  $p_{\mathbf{A}}(\mathbf{A}) = 0$ . This is the whole proof; for general  $n$  there are  $n + 1$  equations and the cancellation is a telescoping sum. The variable  $x$  is never replaced by a matrix: we compare coefficients first, and only then multiply the resulting **matrix** equations by powers of  $\mathbf{A}$ .

*Proof.* Let  $p_{\mathbf{A}} = c_n x^n + c_{n-1} x^{n-1} + \cdots + c_1 x + c_0$  with  $c_n = 1$  (Theorem 6.10.2).

**Step 1: the adjugate as a polynomial with matrix coefficients.** If  $n = 1$ , then  $\text{adj}(x\mathbf{I} - \mathbf{A}) = (1)$  by Definition 6.6.2. If  $n \geq 2$ , each entry of  $\text{adj}(x\mathbf{I} - \mathbf{A})$  is a cofactor  $\pm \det \mathbf{N}$ , where  $\mathbf{N}$  is an  $(n - 1) \times (n - 1)$  submatrix of  $x\mathbf{I} - \mathbf{A}$  (Definition 6.6.2, Definition 6.6.1). Every entry of  $\mathbf{N}$  lies in  $F[x]_{\leq 1}$ . A product of  $n - 1$  such polynomials lies in  $F[x]_{\leq n-1}$  (Theorem 0.8.3, the zero polynomial being harmless), and so does a sum of such products (Theorem 0.8.2); in particular the Leibniz sum  $\det \mathbf{N}$  does (Definition 6.4.1). In both cases every entry of  $\text{adj}(x\mathbf{I} - \mathbf{A})$  lies in  $F[x]_{\leq n-1}$ . For  $0 \leq k \leq n - 1$ , let  $\mathbf{B}_k \in M_n(F)$  be the matrix whose  $(i, j)$ -entry is the coefficient of  $x^k$  in the  $(i, j)$ -entry of  $\text{adj}(x\mathbf{I} - \mathbf{A})$ . Then, entry by entry,

$$\text{adj}(x\mathbf{I} - \mathbf{A}) = \sum_{k=0}^{n-1} x^k \mathbf{B}_k.$$

Put  $\mathbf{B}_{-1} = \mathbf{B}_n = 0$  for convenience.

**Step 2: compare coefficients.** By (1), which holds by Theorem 6.6.2 over the commutative ring  $F[x]$  (the remark after it), the  $(i, j)$ -entries of both sides of (1) agree. On the left, by

Definition 0.9.9,

$$\sum_{l=1}^n (x\delta_{il} - a_{il}) \sum_{k=0}^{n-1} (\mathbf{B}_k)_{lj} x^k = \sum_{k=0}^{n-1} (\mathbf{B}_k)_{ij} x^{k+1} - \sum_{k=0}^{n-1} (\mathbf{A}\mathbf{B}_k)_{ij} x^k,$$

where  $\delta_{il} = 1$  if  $i = l$  and 0 otherwise. So for  $0 \leq k \leq n$ , the coefficient of  $x^k$  is  $(\mathbf{B}_{k-1})_{ij} - (\mathbf{A}\mathbf{B}_k)_{ij}$ . On the right the  $(i, j)$ -entry is  $p_{\mathbf{A}} \delta_{ij}$ , whose coefficient of  $x^k$  is  $c_k \delta_{ij}$ . Two polynomials are equal exactly when their coefficients are equal (Definition 0.8.1). Since this holds for every  $(i, j)$ ,

$$\mathbf{B}_{k-1} - \mathbf{A}\mathbf{B}_k = c_k \mathbf{I} \quad (0 \leq k \leq n). \quad (2)$$

**Step 3: telescope.** Multiply equation (2) for  $k$  on the left by  $\mathbf{A}^k$ , and add the  $n+1$  equations. On the right we get  $\sum_{k=0}^n c_k \mathbf{A}^k = p_{\mathbf{A}}(\mathbf{A})$  by Definition 0.9.14. On the left, writing  $\mathbf{S}_k = \mathbf{A}^k \mathbf{B}_{k-1}$  for  $0 \leq k \leq n+1$ ,

$$\sum_{k=0}^n (\mathbf{A}^k \mathbf{B}_{k-1} - \mathbf{A}^{k+1} \mathbf{B}_k) = \sum_{k=0}^n (\mathbf{S}_k - \mathbf{S}_{k+1}) = \mathbf{S}_0 - \mathbf{S}_{n+1} = \mathbf{B}_{-1} - \mathbf{A}^{n+1} \mathbf{B}_n = 0,$$

since  $\mathbf{B}_{-1} = \mathbf{B}_n = 0$ . Hence  $p_{\mathbf{A}}(\mathbf{A}) = 0$ .

**The operator version.** Let  $\mathcal{B}$  be a basis of  $V$  and  $\mathbf{A} = [T]_{\mathcal{B}}^{\mathcal{B}}$ . By Definition 6.10.2,  $p_T = p_{\mathbf{A}}$ . By Corollary 3.6.3,  $[p_T(T)]_{\mathcal{B}}^{\mathcal{B}} = p_T(\mathbf{A}) = p_{\mathbf{A}}(\mathbf{A}) = 0$ , so  $p_T(T) = 0$  (Theorem 3.6.4). This proves the theorem. ■

Compare with the fake proof. There,  $\mathbf{A}$  was pushed into the entries of  $x\mathbf{I} - \mathbf{A}$ . Here, (1) is first unpacked into the  $n+1$  matrix equations (2), which contain no  $x$  at all, and  $\mathbf{A}$  enters only when we multiply those equations by  $\mathbf{A}^k$ . Every multiplication is on the **left**, so no commutativity of matrices was used; the only commutativity used is inside  $F[x]$ , in the adjugate identity.

Nothing in the proof divides. The determinant, the adjugate identity, coefficient comparison and matrix multiplication all make sense over a commutative ring, and [Exercise \(C1: Cayley–Hamilton over a commutative ring\)](#) asks you to check that the theorem holds, with the same proof, for matrices with entries in a commutative ring such as  $\mathbb{Z}$  or  $\mathbb{R}[t]$ .

The proof also produced more than it was asked for. Equation (2) with  $k = n$  says  $\mathbf{B}_{n-1} = \mathbf{I}$ , and the others say  $\mathbf{B}_{k-1} = \mathbf{A}\mathbf{B}_k + c_k \mathbf{I}$ . Working down from  $\mathbf{B}_{n-1}$ , every  $\mathbf{B}_k$  is a polynomial in  $\mathbf{A}$ ; [Exercise \(C2: The adjugate is a polynomial in the matrix\)](#) turns this into a formula for  $\text{adj } \mathbf{A}$ .

### 8.6.3 Consequences

The first consequence is the promised comparison of the two polynomials.

#### Corollary 8.6.2. The Minimal Polynomial Divides the Characteristic Polynomial

Let  $V$  be finite-dimensional with  $\dim V = n \geq 1$ , and let  $T \in \mathcal{L}(V)$ .

- (a)  $m_T \mid p_T$ ; in particular  $1 \leq \deg m_T \leq n$ .

(b) If  $p_T$  splits over  $F$ , say  $p_T = (x - \lambda_1)^{a_1} \cdots (x - \lambda_k)^{a_k}$  with distinct  $\lambda_i$ , then

$$m_T = (x - \lambda_1)^{s_1} \cdots (x - \lambda_k)^{s_k} \quad \text{with } 1 \leq s_i \leq a_i = a_T(\lambda_i).$$

The same holds for  $\mathbf{A} \in M_n(F)$  with  $m_{\mathbf{A}}$  and  $p_{\mathbf{A}}$ . In particular, over  $\mathbb{C}$ , (b) applies to every  $\mathbf{A} \in M_n(\mathbb{C})$ .

*Proof.*

- (a) By Theorem 8.6.1,  $p_T(T) = 0$ , so  $m_T \mid p_T$  by Theorem 8.5.1. Since  $p_T$  is monic of degree  $n$  (Theorem 6.10.2),  $\deg m_T \leq n$  by Proposition 5.1.2 (c), and  $\deg m_T \geq 1$  by Theorem 8.5.1, as  $V \neq \{0\}$ .
- (b) A polynomial  $x - \lambda$  of degree 1 is irreducible: if  $x - \lambda = gh$ , then  $\deg g + \deg h = 1$  (Theorem 0.8.3), so one factor is constant. The  $x - \lambda_i$  are distinct and monic. By (a) and Lemma 8.5.2,  $m_T = \prod_i (x - \lambda_i)^{s_i}$  with  $0 \leq s_i \leq a_i$ . Each  $\lambda_i$  is a root of  $p_T$ , hence an eigenvalue (Theorem 8.2.1), hence a root of  $m_T$  (Theorem 8.5.3), so  $s_i \geq 1$ . Finally  $a_i = \text{mult}_{\lambda_i}(p_T) = a_T(\lambda_i)$  by Lemma 5.4.1, since the other factors do not vanish at  $\lambda_i$ . The matrix case is the operator  $T_{\mathbf{A}}$ , and over  $\mathbb{C}$  every  $p_{\mathbf{A}}$  splits (Corollary 5.5.5). ■

This pays off a promise from Chapter 5: the bound  $n^2$  of Theorem 5.7.2 improves to  $n$ , and  $p_T$  is an explicit annihilating polynomial of degree exactly  $n$ . It also makes Method 2 of the previous section a finite search that always succeeds. For a matrix with  $p_{\mathbf{A}} = (x - 2)^2(x - 3)$ , the corollary leaves exactly two candidates,  $(x - 2)(x - 3)$  and  $(x - 2)^2(x - 3)$ , and one matrix product decides between them.

**Warning.** Cayley–Hamilton gives an annihilating polynomial, not the minimal one. From  $p_{\mathbf{A}}(\mathbf{A}) = 0$  we may **not** conclude  $m_{\mathbf{A}} = p_{\mathbf{A}}$ ; only  $m_{\mathbf{A}} \mid p_{\mathbf{A}}$ . For  $\mathbf{A} = \text{diag}(1, 1, 2)$ ,  $p_{\mathbf{A}} = (x - 1)^2(x - 2)$  kills  $\mathbf{A}$ , and so does the proper divisor  $(x - 1)(x - 2) = m_{\mathbf{A}}$ . Nor may we conclude that every divisor of  $p_{\mathbf{A}}$  with the right roots kills  $\mathbf{A}$ : for  $\mathbf{J} = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$ ,  $x - 1$  has the right root and does not.

The second consequence expresses inverses through powers.

### Corollary 8.6.3. The Inverse Is a Polynomial in the Matrix

Let  $n \geq 1$ ,  $\mathbf{A} \in M_n(F)$ , and  $p_{\mathbf{A}} = x^n + c_{n-1}x^{n-1} + \cdots + c_1x + c_0$ . Then  $c_0 = (-1)^n \det \mathbf{A}$ . If  $\mathbf{A}$  is invertible, then  $c_0 \neq 0$  and

$$\mathbf{A}^{-1} = -\frac{1}{c_0}(\mathbf{A}^{n-1} + c_{n-1}\mathbf{A}^{n-2} + \cdots + c_2\mathbf{A} + c_1\mathbf{I}).$$

(For  $n = 1$  the bracket is  $\mathbf{I}$  alone, and the formula reads  $\mathbf{A}^{-1} = -c_0^{-1}\mathbf{I}$ .) The same holds for an invertible operator  $T$  on a space of dimension  $n \geq 1$ , with  $\text{id}_V$  in place of  $\mathbf{I}$ .

*Proof.* By Theorem 6.10.2,  $c_0 = (-1)^n \det \mathbf{A}$ , which is non-zero when  $\mathbf{A}$  is invertible (Theorem 6.5.3). By Theorem 8.6.1,

$$\mathbf{A}(\mathbf{A}^{n-1} + c_{n-1}\mathbf{A}^{n-2} + \cdots + c_1\mathbf{I}) = \mathbf{A}^n + c_{n-1}\mathbf{A}^{n-1} + \cdots + c_1\mathbf{A} = -c_0\mathbf{I}.$$

Multiplying by  $-c_0^{-1}$ , the matrix in parentheses times  $-c_0^{-1}$  is a right inverse of  $\mathbf{A}$ , hence the inverse by Theorem 2.4.5. For an operator, apply this to  $[T]_{\mathcal{B}}^{\mathcal{B}}$  and use Corollary 3.6.3. ■

So inverting an  $n \times n$  matrix never requires more than its first  $n - 1$  powers and the coefficients of  $p_{\mathbf{A}}$ . The theorem also reduces **every** power of  $\mathbf{A}$  to the first  $n - 1$ . Divide  $x^N$  by  $p_{\mathbf{A}}$  (Theorem 5.1.1):  $x^N = qp_{\mathbf{A}} + r$  with  $\deg r < n$ . Then by Theorem 5.7.1 and Theorem 8.6.1,

$$\mathbf{A}^N = q(\mathbf{A})p_{\mathbf{A}}(\mathbf{A}) + r(\mathbf{A}) = r(\mathbf{A}).$$

When  $p_{\mathbf{A}}$  has  $n$  distinct roots  $\lambda_1, \dots, \lambda_n$ , the remainder can be found without long division. Substituting  $x = \lambda_i$  in  $x^N = qp_{\mathbf{A}} + r$  gives  $r(\lambda_i) = \lambda_i^N$  for each  $i$ , and these  $n$  conditions determine  $r \in F[x]_{\leq n-1}$  by Lagrange interpolation (Theorem 5.6.1). This substitution is legal:  $\lambda_i$  is a **scalar** plugged into an identity of polynomials.

#### Example. Cayley–Hamilton for a $3 \times 3$ Matrix

Let

$$\mathbf{A} = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 1 \end{pmatrix} \in M_3(\mathbb{Q}).$$

Compute  $p_{\mathbf{A}}$  and verify  $p_{\mathbf{A}}(\mathbf{A}) = 0$ . Then use it to find  $\mathbf{A}^{-1}$  and  $\mathbf{A}^5$ , and write down the matrices  $\mathbf{B}_k$  of the proof.

*Solution. Characteristic polynomial.* Expand  $\det(x\mathbf{I} - \mathbf{A})$  along the first row (Theorem 6.6.1 over  $\mathbb{Q}[x]$ ):

$$\begin{aligned} p_{\mathbf{A}} &= \det \begin{pmatrix} x-1 & -1 & 0 \\ 0 & x & -1 \\ -1 & 0 & x-1 \end{pmatrix} \\ &= (x-1) \det \begin{pmatrix} x & -1 \\ 0 & x-1 \end{pmatrix} + (-1)(-1)^{1+2} \det \begin{pmatrix} 0 & -1 \\ -1 & x-1 \end{pmatrix}. \end{aligned}$$

The two minors are  $x(x-1)$  and  $-1$ , so  $p_{\mathbf{A}} = x(x-1)^2 - 1 = x^3 - 2x^2 + x - 1$ . As a check,  $\text{tr } \mathbf{A} = 2$  matches the coefficient  $-2$  of  $x^2$ , and  $c_0 = -1 = (-1)^3 \det \mathbf{A}$  says  $\det \mathbf{A} = 1$ .

*Verification.* Multiplying out,

$$\mathbf{A}^2 = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 0 & 1 \\ 2 & 1 & 1 \end{pmatrix}, \quad \mathbf{A}^3 = \mathbf{A}\mathbf{A}^2 = \begin{pmatrix} 2 & 1 & 2 \\ 2 & 1 & 1 \\ 3 & 2 & 2 \end{pmatrix}.$$

Then  $\mathbf{A}^3 - 2\mathbf{A}^2 + \mathbf{A} - \mathbf{I}$  has first row  $(2 - 2 + 1 - 1, 1 - 2 + 1, 2 - 2 + 0) = (0, 0, 0)$ , second row

$(2-2+0, 1-0+0-1, 1-2+1) = (0, 0, 0)$ , and third row  $(3-4+1, 2-2+0, 2-2+1-1) = (0, 0, 0)$ . So  $p_{\mathbf{A}}(\mathbf{A}) = 0$ , as the theorem predicts.

*Inverse.* Here  $c_0 = -1$ ,  $c_1 = 1$ ,  $c_2 = -2$ . By [Corollary 8.6.3](#),

$$\mathbf{A}^{-1} = \mathbf{A}^2 - 2\mathbf{A} + \mathbf{I} = \begin{pmatrix} 0 & -1 & 1 \\ 1 & 1 & -1 \\ 0 & 1 & 0 \end{pmatrix}.$$

Check: the first row of  $\mathbf{A} \mathbf{A}^{-1}$  is  $(0+1, -1+1, 1-1) = (1, 0, 0)$ , and the other rows work the same way. The entries are integers, as they must be for an integer matrix of determinant 1 ([Corollary 6.6.5](#)).

*Fifth power.* The relation  $\mathbf{A}^3 = 2\mathbf{A}^2 - \mathbf{A} + \mathbf{I}$  lets us lower degrees step by step:

$$\mathbf{A}^4 = 2\mathbf{A}^3 - \mathbf{A}^2 + \mathbf{A} = 2(2\mathbf{A}^2 - \mathbf{A} + \mathbf{I}) - \mathbf{A}^2 + \mathbf{A} = 3\mathbf{A}^2 - \mathbf{A} + 2\mathbf{I},$$

$$\mathbf{A}^5 = 3\mathbf{A}^3 - \mathbf{A}^2 + 2\mathbf{A} = 3(2\mathbf{A}^2 - \mathbf{A} + \mathbf{I}) - \mathbf{A}^2 + 2\mathbf{A} = 5\mathbf{A}^2 - \mathbf{A} + 3\mathbf{I}.$$

(Equivalently,  $x^5 = (x^2 + 2x + 3)p_{\mathbf{A}} + (5x^2 - x + 3)$ .) Hence

$$\mathbf{A}^5 = 5 \begin{pmatrix} 1 & 1 & 1 \\ 1 & 0 & 1 \\ 2 & 1 & 1 \end{pmatrix} - \begin{pmatrix} 1 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 1 \end{pmatrix} + 3\mathbf{I} = \begin{pmatrix} 7 & 4 & 5 \\ 5 & 3 & 4 \\ 9 & 5 & 7 \end{pmatrix}.$$

*The matrices of the proof.* By (2),  $\mathbf{B}_2 = \mathbf{I}$ ,  $\mathbf{B}_1 = \mathbf{A}\mathbf{B}_2 + c_2\mathbf{I} = \mathbf{A} - 2\mathbf{I}$  and  $\mathbf{B}_0 = \mathbf{A}\mathbf{B}_1 + c_1\mathbf{I} = \mathbf{A}^2 - 2\mathbf{A} + \mathbf{I}$ . So

$$\begin{aligned} \text{adj}(x\mathbf{I} - \mathbf{A}) &= x^2\mathbf{I} + x(\mathbf{A} - 2\mathbf{I}) + (\mathbf{A}^2 - 2\mathbf{A} + \mathbf{I}) \\ &= \begin{pmatrix} x^2 - x & x - 1 & 1 \\ 1 & x^2 - 2x + 1 & x - 1 \\ x & 1 & x^2 - x \end{pmatrix}, \end{aligned}$$

which one can confirm entry by entry from cofactors of  $x\mathbf{I} - \mathbf{A}$ ; for instance the  $(1, 3)$ -entry is the cofactor  $C_{31} = \det \begin{pmatrix} -1 & 0 \\ x & -1 \end{pmatrix} = 1$ . The last equation of (2),  $-\mathbf{A}\mathbf{B}_0 = c_0\mathbf{I} = -\mathbf{I}$ , is the statement  $\mathbf{A}^{-1} = \mathbf{B}_0$  found above.

*Quick check.* Let  $\mathbf{A} \in M_2(\mathbb{R})$  with  $\text{tr } \mathbf{A} = 5$  and  $\det \mathbf{A} = 6$ . Write  $\mathbf{A}^2$  and  $\mathbf{A}^3$  as combinations of  $\mathbf{A}$  and  $\mathbf{I}$ , and write  $\mathbf{A}^{-1}$  the same way.

*Solution.*  $p_{\mathbf{A}} = x^2 - 5x + 6$ , so by [Theorem 8.6.1](#)  $\mathbf{A}^2 = 5\mathbf{A} - 6\mathbf{I}$ . Then  $\mathbf{A}^3 = 5\mathbf{A}^2 - 6\mathbf{A} = 5(5\mathbf{A} - 6\mathbf{I}) - 6\mathbf{A} = 19\mathbf{A} - 30\mathbf{I}$ . By [Corollary 8.6.3](#) with  $c_0 = 6$  and  $c_1 = -5$ ,  $\mathbf{A}^{-1} = -\frac{1}{6}(\mathbf{A} - 5\mathbf{I}) = \frac{1}{6}(5\mathbf{I} - \mathbf{A})$ .

**Alternatively.** When  $p_T$  splits, there is a second proof that avoids determinants of polynomial matrices entirely. The next section shows that such an operator has an upper triangular matrix, and a triangular matrix reveals directly why the product  $(T - \lambda_1 \text{id}_V) \cdots (T - \lambda_n \text{id}_V)$  is zero. That proof is set out as an exercise there. The adjugate proof has the advantage of working

over every field at once, with no splitting hypothesis.

## 8.6.4 Exercises

### 8.6.4.1 A. Check your understanding

#### Exercise. A1

- State the Cayley–Hamilton Theorem for  $\mathbf{A} \in M_n(F)$ .
- True or false:  $\deg m_{\mathbf{A}} \leq n$  for every  $\mathbf{A} \in M_n(F)$ ,  $n \geq 1$ . Justify your answer.
- In one sentence, why is “ $p_{\mathbf{A}}(\mathbf{A}) = \det(\mathbf{A}\mathbf{I} - \mathbf{A}) = 0$ ” not a proof?
- Suppose  $p_{\mathbf{A}} = (x - 1)^2(x + 2)$ . Which polynomials can  $m_{\mathbf{A}}$  be?
- True or false: if  $p_{\mathbf{A}}(\mathbf{A}) = 0$ , then  $p_{\mathbf{A}}$  is the minimal polynomial of  $\mathbf{A}$ . Justify your answer.
- How do you write  $\mathbf{A}^{-1}$  as a polynomial in  $\mathbf{A}$  when  $\mathbf{A}$  is invertible?

*Solution.*

- If  $n \geq 1$  and  $\mathbf{A} \in M_n(F)$ , then  $p_{\mathbf{A}}(\mathbf{A}) = 0$  (Theorem 8.6.1).
- True. By Corollary 8.6.2,  $m_{\mathbf{A}} \mid p_{\mathbf{A}}$ , and  $\deg p_{\mathbf{A}} = n$ .
- It substitutes a matrix into the entries of  $x\mathbf{I} - \mathbf{A}$  and then equates the matrix  $p_{\mathbf{A}}(\mathbf{A})$  with the scalar  $\det 0$ , neither of which is legitimate.
- By Corollary 8.6.2 (b),  $m_{\mathbf{A}} = (x - 1)^s(x + 2)^t$  with  $1 \leq s \leq 2$  and  $t = 1$ : so  $(x - 1)(x + 2)$  or  $(x - 1)^2(x + 2)$ . Both occur, for  $\text{diag}(1, 1, -2)$  and for  $\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} \oplus (-2)$  (Proposition 8.5.6).
- False.  $p_{\mathbf{I}_2} = (x - 1)^2$  kills  $\mathbf{I}_2$ , but  $m_{\mathbf{I}_2} = x - 1$ .
- If  $p_{\mathbf{A}} = x^n + c_{n-1}x^{n-1} + \cdots + c_0$ , then  $c_0 = (-1)^n \det \mathbf{A} \neq 0$  and  $\mathbf{A}^{-1} = -c_0^{-1}(\mathbf{A}^{n-1} + c_{n-1}\mathbf{A}^{n-2} + \cdots + c_1\mathbf{I})$  (Corollary 8.6.3).

### 8.6.4.2 B. Practice

#### Exercise. B1: Inverse and a hundredth power

Let  $\mathbf{A} = \begin{pmatrix} 3 & -2 \\ 1 & 0 \end{pmatrix} \in M_2(\mathbb{R})$ .

- Find  $p_{\mathbf{A}}$ , and use the Cayley–Hamilton Theorem to compute  $\mathbf{A}^{-1}$ .
- Hence find  $\mathbf{A}^{100}$  as a combination  $a\mathbf{A} + b\mathbf{I}$ , and write it as a matrix.

*Solution.*

- $\text{tr } \mathbf{A} = 3$  and  $\det \mathbf{A} = 2$ , so  $p_{\mathbf{A}} = x^2 - 3x + 2 = (x - 1)(x - 2)$ . By Theorem 8.6.1,

$\mathbf{A}^2 - 3\mathbf{A} + 2\mathbf{I} = 0$ , that is,  $\mathbf{A}(3\mathbf{I} - \mathbf{A}) = 2\mathbf{I}$ . By [Corollary 8.6.3](#),

$$\mathbf{A}^{-1} = \frac{1}{2}(3\mathbf{I} - \mathbf{A}) = \begin{pmatrix} 0 & 1 \\ -\frac{1}{2} & \frac{3}{2} \end{pmatrix}.$$

Check:  $\mathbf{A}\mathbf{A}^{-1} = \begin{pmatrix} 0+1 & 3-3 \\ 0 & 1 \end{pmatrix} = \mathbf{I}$ .

- (b) By Theorem 5.1.1,  $x^{100} = qp_{\mathbf{A}} + (ax + b)$  for some  $q$  and scalars  $a, b$ . Substituting the roots  $x = 1$  and  $x = 2$  of  $p_{\mathbf{A}}$  gives  $1 = a + b$  and  $2^{100} = 2a + b$ . So  $a = 2^{100} - 1$  and  $b = 2 - 2^{100}$ . By [Theorem 8.6.1](#),  $\mathbf{A}^{100} = q(\mathbf{A})p_{\mathbf{A}}(\mathbf{A}) + a\mathbf{A} + b\mathbf{I} = a\mathbf{A} + b\mathbf{I}$ , that is,

$$\mathbf{A}^{100} = (2^{100} - 1)\mathbf{A} + (2 - 2^{100})\mathbf{I} = \begin{pmatrix} 2^{101} - 1 & 2 - 2^{101} \\ 2^{100} - 1 & 2 - 2^{100} \end{pmatrix}.$$

For  $k = 1$  in place of 100, the same recipe gives  $a = 1$ ,  $b = 0$ , and returns  $\mathbf{A}$ , as it should.

### Exercise. B2: Trace zero in size two

- (a) Let  $\mathbf{A} \in M_2(F)$  with  $\text{tr } \mathbf{A} = 0$ . Prove that  $\mathbf{A}^2 = -(\det \mathbf{A})\mathbf{I}$ .  
 (b) Deduce that for all  $\mathbf{X}, \mathbf{Y} \in M_2(F)$ , the matrix  $(\mathbf{XY} - \mathbf{YX})^2$  is a scalar multiple of  $\mathbf{I}$ .  
 (c) Show that the converse of (a) fails: give  $\mathbf{A} \in M_2(\mathbb{R})$  with  $\mathbf{A}^2$  scalar and  $\text{tr } \mathbf{A} \neq 0$ .

*Solution.*

- (a) By [Theorem 8.6.1](#),  $\mathbf{A}^2 - (\text{tr } \mathbf{A})\mathbf{A} + (\det \mathbf{A})\mathbf{I} = 0$ . With  $\text{tr } \mathbf{A} = 0$ , this is  $\mathbf{A}^2 = -(\det \mathbf{A})\mathbf{I}$ .  
 (b) By Theorem 3.10.3 applied to  $T_{\mathbf{X}}, T_{\mathbf{Y}}$ , or directly by  $\text{tr}(\mathbf{XY}) = \text{tr}(\mathbf{YX})$ , we have  $\text{tr}(\mathbf{XY} - \mathbf{YX}) = 0$ . By (a),  $(\mathbf{XY} - \mathbf{YX})^2 = -\det(\mathbf{XY} - \mathbf{YX})\mathbf{I}$ .  
 (c)  $\mathbf{A} = \mathbf{I}_2$  has  $\mathbf{A}^2 = \mathbf{I}_2$  and  $\text{tr } \mathbf{A} = 2 \neq 0$ .

### Exercise. B3: Minimal polynomial from one product

Each of the real matrices

$$\mathbf{M}_1 = \begin{pmatrix} 2 & 1 & 0 \\ 0 & 2 & 0 \\ -1 & 1 & 3 \end{pmatrix}, \quad \mathbf{M}_2 = \begin{pmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ -1 & 0 & 3 \end{pmatrix}$$

has characteristic polynomial  $(x-2)^2(x-3)$ . Find  $m_{\mathbf{M}_1}$  and  $m_{\mathbf{M}_2}$ . Are  $\mathbf{M}_1$  and  $\mathbf{M}_2$  similar?

*Solution.* By [Corollary 8.6.2](#) (b), each minimal polynomial is  $(x-2)(x-3)$  or  $(x-2)^2(x-3)$ , and only the first needs testing.

For  $\mathbf{M}_1$ :  $\mathbf{M}_1 - 2\mathbf{I} = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 0 \\ -1 & 1 & 1 \end{pmatrix}$  and  $\mathbf{M}_1 - 3\mathbf{I} = \begin{pmatrix} -1 & 1 & 0 \\ 0 & -1 & 0 \\ -1 & 1 & 0 \end{pmatrix}$ . The first row of the product

$(\mathbf{M}_1 - 2\mathbf{I})(\mathbf{M}_1 - 3\mathbf{I})$  is row 2 of  $\mathbf{M}_1 - 3\mathbf{I}$ , namely  $(0, -1, 0) \neq \mathbf{0}$ . So  $m_{\mathbf{M}_1} = (x - 2)^2(x - 3)$ .

For  $\mathbf{M}_2$ :  $\mathbf{M}_2 - 2\mathbf{I} = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ -1 & 0 & 1 \end{pmatrix}$  and  $\mathbf{M}_2 - 3\mathbf{I} = \begin{pmatrix} -1 & 0 & 0 \\ 0 & -1 & 0 \\ -1 & 0 & 0 \end{pmatrix}$ . The only non-zero row of the product is  $-(\text{row } 1) + (\text{row } 3)$  of  $\mathbf{M}_2 - 3\mathbf{I}$ , which is  $(1, 0, 0) + (-1, 0, 0) = \mathbf{0}$ . So the product is 0, and  $m_{\mathbf{M}_2} = (x - 2)(x - 3)$ .

The minimal polynomials differ, so  $\mathbf{M}_1 \not\sim \mathbf{M}_2$  by Theorem 8.5.5.

### 8.6.4.3 C. Going deeper

#### Exercise. C1: Cayley–Hamilton over a commutative ring

Let  $R$  be a commutative ring (the remark in Chapter 6, §4), and suppose, as for  $F[x]$ , that the polynomials in  $x$  with coefficients in  $R$  form a commutative ring  $R[x]$  in which two polynomials are equal exactly when all their coefficients are equal. For  $\mathbf{A} \in M_n(R)$ , define  $p_{\mathbf{A}} = \det(x\mathbf{I} - \mathbf{A}) \in R[x]$ .

- Go through the proof of Theorem 8.6.1 and name, for each step, the fact it uses. Explain why each fact holds over  $R$  and  $R[x]$ , and conclude that  $p_{\mathbf{A}}(\mathbf{A}) = 0$  for every  $\mathbf{A} \in M_n(R)$ .
- Let  $R = \mathbb{R}[t]$  and  $\mathbf{A} = \begin{pmatrix} t & 1 \\ t^2 & 0 \end{pmatrix}$ . Compute  $p_{\mathbf{A}}$  and verify  $p_{\mathbf{A}}(\mathbf{A}) = 0$  directly.
- Let  $\mathbf{A} \in M_n(\mathbb{Z})$  with  $\det \mathbf{A} = \pm 1$ . Using (a) with  $R = \mathbb{Z}$ , show that  $\mathbf{A}^{-1}$  has integer entries.

*Solution.*

- The shape of  $p_{\mathbf{A}}$ .* In the Leibniz sum for  $\det(x\mathbf{I} - \mathbf{A})$ , the identity permutation contributes  $(x - a_{11}) \cdots (x - a_{nn})$ , of degree  $n$  with leading coefficient 1, and every other term omits at least two diagonal entries, so it has degree at most  $n - 2$ . Hence  $p_{\mathbf{A}} = x^n + c_{n-1}x^{n-1} + \cdots + c_0$  with  $c_k \in R$ , which is what the proof uses in place of Theorem 6.10.2; only sums and products of entries appear. *Step 1* uses the Leibniz formula over  $R[x]$  (which only multiplies and adds entries) and the degree bounds for sums and products, which hold because the degree of a product of polynomials is at most the sum of the degrees; no inverse is needed for the upper bound. *Step 2* uses the adjugate identity over the commutative ring  $R[x]$ , which holds by the remark after Theorem 6.6.2, and coefficient comparison in  $R[x]$ , which is the assumption on  $R[x]$ . *Step 3* uses only associativity and distributivity of matrix multiplication over  $R$  and a telescoping sum. No step divides, so each works verbatim over  $R$ , and  $p_{\mathbf{A}}(\mathbf{A}) = 0$ .

- $x\mathbf{I} - \mathbf{A} = \begin{pmatrix} x - t & -1 \\ -t^2 & x \end{pmatrix}$ , so  $p_{\mathbf{A}} = x(x - t) - t^2 = x^2 - tx - t^2$ . Then

$$\mathbf{A}^2 = \begin{pmatrix} t^2 + t^2 & t \\ t^3 & t^2 \end{pmatrix}, \quad \mathbf{A}^2 - t\mathbf{A} - t^2\mathbf{I} = \begin{pmatrix} 2t^2 - t^2 - t^2 & t - t \\ t^3 - t^3 & t^2 - 0 - t^2 \end{pmatrix} = \mathbf{0}.$$

- The Leibniz formula for  $\det(x\mathbf{I} - \mathbf{A})$  only adds and multiplies integers and  $x$ , so the coefficients  $c_k$  of  $p_{\mathbf{A}}$  are integers. Regarding  $\mathbf{A}$  as a rational matrix, Theorem 6.10.2 gives  $c_0 = (-1)^n \det \mathbf{A} = \pm 1$ . By (a),  $\mathbf{A}(\mathbf{A}^{n-1} + c_{n-1}\mathbf{A}^{n-2} + \cdots + c_1\mathbf{I}) = -c_0\mathbf{I}$ . Since

$c_0^2 = 1$ , multiplying by  $-c_0$  gives  $\mathbf{A}\mathbf{B} = \mathbf{I}$  with  $\mathbf{B} = -c_0(\mathbf{A}^{n-1} + \cdots + c_1\mathbf{I}) \in M_n(\mathbb{Z})$ . Regarding  $\mathbf{A}$  and  $\mathbf{B}$  as rational matrices, Theorem 2.4.5 gives  $\mathbf{B} = \mathbf{A}^{-1}$ , which therefore has integer entries. (This reproves part of Corollary 6.6.5.)

**Exercise. C2: The adjugate is a polynomial in the matrix**

Let  $n \geq 2$ ,  $\mathbf{A} \in M_n(F)$ ,  $p_{\mathbf{A}} = x^n + c_{n-1}x^{n-1} + \cdots + c_0$ , and let  $\mathbf{B}_0, \dots, \mathbf{B}_{n-1}$  be as in the proof of Theorem 8.6.1.

- (a) Using equation (2) of the proof, show that  $\mathbf{B}_0 = \mathbf{A}^{n-1} + c_{n-1}\mathbf{A}^{n-2} + \cdots + c_2\mathbf{A} + c_1\mathbf{I}$ .
- (b) Show that the constant matrix  $\mathbf{B}_0$  equals  $\text{adj}(-\mathbf{A})$ , and that  $\text{adj}(-\mathbf{A}) = (-1)^{n-1} \text{adj } \mathbf{A}$ .
- (c) Deduce that for **every**  $\mathbf{A} \in M_n(F)$ , invertible or not,

$$\text{adj } \mathbf{A} = (-1)^{n-1}(\mathbf{A}^{n-1} + c_{n-1}\mathbf{A}^{n-2} + \cdots + c_2\mathbf{A} + c_1\mathbf{I}).$$

- (d) Check (c) for  $\mathbf{S} = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 1 & 0 & 1 \end{pmatrix}$ , whose adjugate Chapter 6 computed (the warning after Corollary 6.6.3), given that  $p_{\mathbf{S}} = x^3 - 6x^2 + 2x$ .

*Solution.*

- (a) Equation (2) with  $k = n$  gives  $\mathbf{B}_{n-1} = c_n\mathbf{I} = \mathbf{I}$ , and for  $1 \leq k \leq n-1$  it gives  $\mathbf{B}_{k-1} = \mathbf{A}\mathbf{B}_k + c_k\mathbf{I}$ . By downward induction on  $k$ ,  $\mathbf{B}_{k-1} = \mathbf{A}^{n-k} + c_{n-1}\mathbf{A}^{n-k-1} + \cdots + c_k\mathbf{I}$ : this holds for  $k = n$ , and if it holds for  $k$ , then  $\mathbf{B}_{k-2} = \mathbf{A}\mathbf{B}_{k-1} + c_{k-1}\mathbf{I} = \mathbf{A}^{n-k+1} + c_{n-1}\mathbf{A}^{n-k} + \cdots + c_k\mathbf{A} + c_{k-1}\mathbf{I}$ . The case  $k = 1$  is the claim.
- (b) Each entry of  $\text{adj}(x\mathbf{I} - \mathbf{A})$  is a cofactor  $\pm \det \mathbf{N}$  of  $x\mathbf{I} - \mathbf{A}$ , and its constant coefficient is its value at 0. By Lemma 7.3.7,  $(\det \mathbf{N})(0) = \det \mathbf{N}(0)$ , and  $\mathbf{N}(0)$  is the corresponding submatrix of  $-\mathbf{A}$ . So the matrix of constant coefficients,  $\mathbf{B}_0$ , is  $\text{adj}(-\mathbf{A})$ . Each cofactor of  $-\mathbf{A}$  is  $\pm$  the determinant of an  $(n-1) \times (n-1)$  submatrix of  $-\mathbf{A}$ , which is  $-1$  times the corresponding submatrix of  $\mathbf{A}$ . Multiplying each of the  $n-1$  rows by  $-1$  multiplies the determinant by  $(-1)^{n-1}$  (Theorem 6.5.1). Hence  $\text{adj}(-\mathbf{A}) = (-1)^{n-1} \text{adj } \mathbf{A}$ .
- (c) By (a) and (b),  $(-1)^{n-1} \text{adj } \mathbf{A} = \mathbf{B}_0 = \mathbf{A}^{n-1} + c_{n-1}\mathbf{A}^{n-2} + \cdots + c_1\mathbf{I}$ ; multiply by  $(-1)^{n-1}$ . Nothing assumed  $\det \mathbf{A} \neq 0$ .
- (d) Here  $n = 3$ ,  $c_2 = -6$ ,  $c_1 = 2$ , so the formula predicts  $\text{adj } \mathbf{S} = \mathbf{S}^2 - 6\mathbf{S} + 2\mathbf{I}$ . Now

$$\begin{aligned} \mathbf{S}^2 &= \begin{pmatrix} 8 & 10 & 18 \\ 16 & 20 & 36 \\ 2 & 2 & 4 \end{pmatrix}, \\ \mathbf{S}^2 - 6\mathbf{S} + 2\mathbf{I} &= \begin{pmatrix} 8-6+2 & 10-12 & 18-18 \\ 16-12 & 20-24+2 & 36-36 \\ 2-6 & 2 & 4-6+2 \end{pmatrix} \\ &= \begin{pmatrix} 4 & -2 & 0 \\ 4 & -2 & 0 \\ -4 & 2 & 0 \end{pmatrix}, \end{aligned}$$

which is the adjugate found in Chapter 6.

**Exercise. C3: Nilpotent matrices of size  $n$** 

Let  $F$  be any field,  $n \geq 1$ , and  $\mathbf{A} \in M_n(F)$  with  $\mathbf{A}^k = 0$  for some  $k \geq 1$ .

- (a) Prove that  $m_{\mathbf{A}} = x^s$  for some  $s$  with  $1 \leq s \leq n$ .
- (b) Deduce that  $\mathbf{A}^n = 0$ .
- (c) Give a  $3 \times 3$  example with  $\mathbf{A}^2 \neq 0$ , and explain why no  $3 \times 3$  matrix has  $\mathbf{A}^3 \neq 0 = \mathbf{A}^4$ .

*Solution.*

- (a)  $x^k$  annihilates  $\mathbf{A}$ , so  $m_{\mathbf{A}} \mid x^k$  (Theorem 8.5.1). The polynomial  $x$  is monic and irreducible (degree 1), so by Lemma 8.5.2,  $m_{\mathbf{A}} = x^s$  with  $0 \leq s \leq k$ . By [Corollary 8.6.2](#) (a),  $1 \leq s = \deg m_{\mathbf{A}} \leq n$ .
- (b)  $x^n = x^{n-s} x^s$ , so  $m_{\mathbf{A}} \mid x^n$  and  $\mathbf{A}^n = 0$  by Theorem 8.5.1.
- (c) The matrix  $\mathbf{N}$  of the Quick check in Section 5 has  $\mathbf{N}^2 \neq 0 = \mathbf{N}^3$ . If  $\mathbf{A} \in M_3(F)$  satisfies  $\mathbf{A}^4 = 0$ , then  $\mathbf{A}$  is nilpotent and  $\mathbf{A}^3 = 0$  by (b), so  $\mathbf{A}^3 \neq 0 = \mathbf{A}^4$  is impossible. Note that no splitting or complex numbers were needed; compare Exercise (C2: Nilpotent operators), which worked over  $\mathbb{C}$ .